

BABU BANARASI DAS UNIVERSITY

2026–2027



SCHOOL OF COMPUTER APPLICATIONS

NoSQL Project File

Student Management System Using MongoDB

Submitted To:

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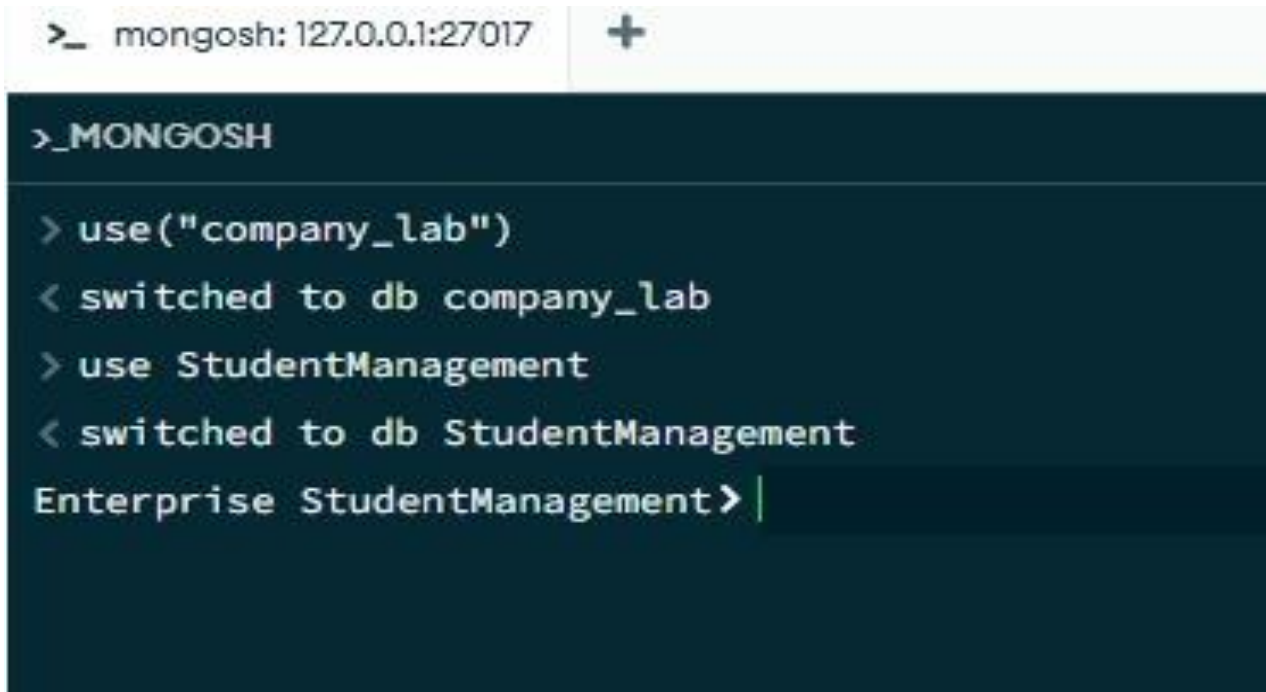
Roll no : 1250258082

1. CREATE DATABASE

MongoDB Query:

```
use StudentManagement
```

Explanation: Creates or switches to the StudentManagement database.



The screenshot shows a MongoDB shell interface. At the top, a tab is labeled '>_ mongosh: 127.0.0.1:27017' with a plus icon to its right. The main terminal area has a dark background with light green text. It starts with '>_MONGOSH'. The first command is '> use("company_lab")', followed by the response '< switched to db company_lab'. The second command is '> use StudentManagement', followed by the response '< switched to db StudentManagement'. The prompt 'Enterprise StudentManagement>' is shown with a vertical cursor bar at the end.

2. CREATE COLLECTION

MongoDB Query:

```
db.createCollection("students")
```

Explanation: Creates a collection named students.

▼ 127.0.0.1:27017

▼ StudentManagement

students

▼ admin

▶ company_lab

▶ config

▶ local

▶ test

Enterpr

3. INSERT 50 STUDENTS

MongoDB Query:

```
db.students.insertMany([
  {roll:1, name:"Aarav", age:19, marks:88, city:"Delhi"},
  {roll:2, name:"Aditi", age:19, marks:69, city:"Ghaziabad"},
  {roll:3, name:"Arjun", age:18, marks:89, city:"Noida"},
  {roll:4, name:"Bhavya", age:21, marks:57, city:"Delhi"},
```

NoSQL Project – Student Management System Using MongoDB

```

{roll:5, name:"Chirag", age:19, marks:78, city:"Noida"},
{roll:6, name:"Diya", age:18, marks:90, city:"Lucknow"},
{roll:7, name:"Dev", age:21, marks:69, city:"Varanasi"},
{roll:8, name:"Ishita", age:20, marks:55, city:"Ghaziabad"},
{roll:9, name:"Kunal", age:21, marks:76, city:"Kanpur"},
{roll:10, name:"Mehak", age:18, marks:64, city:"Delhi"},
{roll:11, name:"Naman", age:18, marks:60, city:"Prayagraj"},
{roll:12, name:"Pallavi", age:18, marks:77, city:"Agra"},
{roll:13, name:"Ritik", age:20, marks:57, city:"Varanasi"},
{roll:14, name:"Simran", age:18, marks:79, city:"Noida"},
{roll:15, name:"Tanvi", age:20, marks:95, city:"Bareilly"},
{roll:16, name:"Varun", age:20, marks:91, city:"Lucknow"},
{roll:17, name:"Kavya", age:18, marks:57, city:"Lucknow"},
{roll:18, name:"Raghav", age:20, marks:60, city:"Lucknow"},
{roll:19, name:"Muskan", age:18, marks:79, city:"Kanpur"},
{roll:20, name:"Siddharth", age:21, marks:95, city:"Agra"},
{roll:21, name:"Nikita", age:19, marks:78, city:"Agra"},
{roll:22, name:"Yash", age:19, marks:97, city:"Kanpur"},
{roll:23, name:"Ishaan", age:18, marks:93, city:"Ghaziabad"},
{roll:24, name:"Riya", age:19, marks:65, city:"Varanasi"},
{roll:25, name:"Mohit", age:21, marks:72, city:"Meerut"},
{roll:26, name:"Sakshi", age:19, marks:75, city:"Delhi"},
{roll:27, name:"Ankit", age:19, marks:57, city:"Agra"},
{roll:28, name:"Pooja", age:21, marks:72, city:"Noida"},
{roll:29, name:"Rahul", age:19, marks:91, city:"Agra"},
{roll:30, name:"Sneha", age:19, marks:96, city:"Varanasi"},
{roll:31, name:"Manish", age:21, marks:96, city:"Varanasi"},
{roll:32, name:"Komal", age:19, marks:71, city:"Ghaziabad"},
{roll:33, name:"Abhishek", age:19, marks:90, city:"Meerut"},
{roll:34, name:"Nandini", age:20, marks:92, city:"Prayagraj"},
{roll:35, name:"Rohan", age:21, marks:78, city:"Lucknow"},
{roll:36, name:"Shivani", age:19, marks:87, city:"Varanasi"},
{roll:37, name:"Harshit", age:18, marks:58, city:"Noida"},
{roll:38, name:"Preeti", age:19, marks:95, city:"Ghaziabad"},
{roll:39, name:"Akash", age:21, marks:93, city:"Noida"},
{roll:40, name:"Anjali", age:21, marks:79, city:"Bareilly"},
{roll:41, name:"Vivek", age:21, marks:88, city:"Kanpur"},
{roll:42, name:"Neha", age:18, marks:62, city:"Meerut"},
{roll:43, name:"Tushar", age:20, marks:96, city:"Agra"},
{roll:44, name:"Shreya", age:18, marks:73, city:"Prayagraj"},
{roll:45, name:"Aditya", age:19, marks:84, city:"Delhi"},
{roll:46, name:"Priyanka", age:20, marks:87, city:"Ghaziabad"},
{roll:47, name:"Ayush", age:18, marks:95, city:"Kanpur"},
{roll:48, name:"Mansi", age:19, marks:64, city:"Agra"},
{roll:49, name:"Dhruv", age:19, marks:89, city:"Meerut"},
{roll:50, name:"Anushka", age:20, marks:91, city:"Lucknow"}
]

```

Explanation: Inserts 50 different student documents into the students collection.

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

127.0.0.1:27017

StudentManagement

students

admin

company_lab

config

local

test

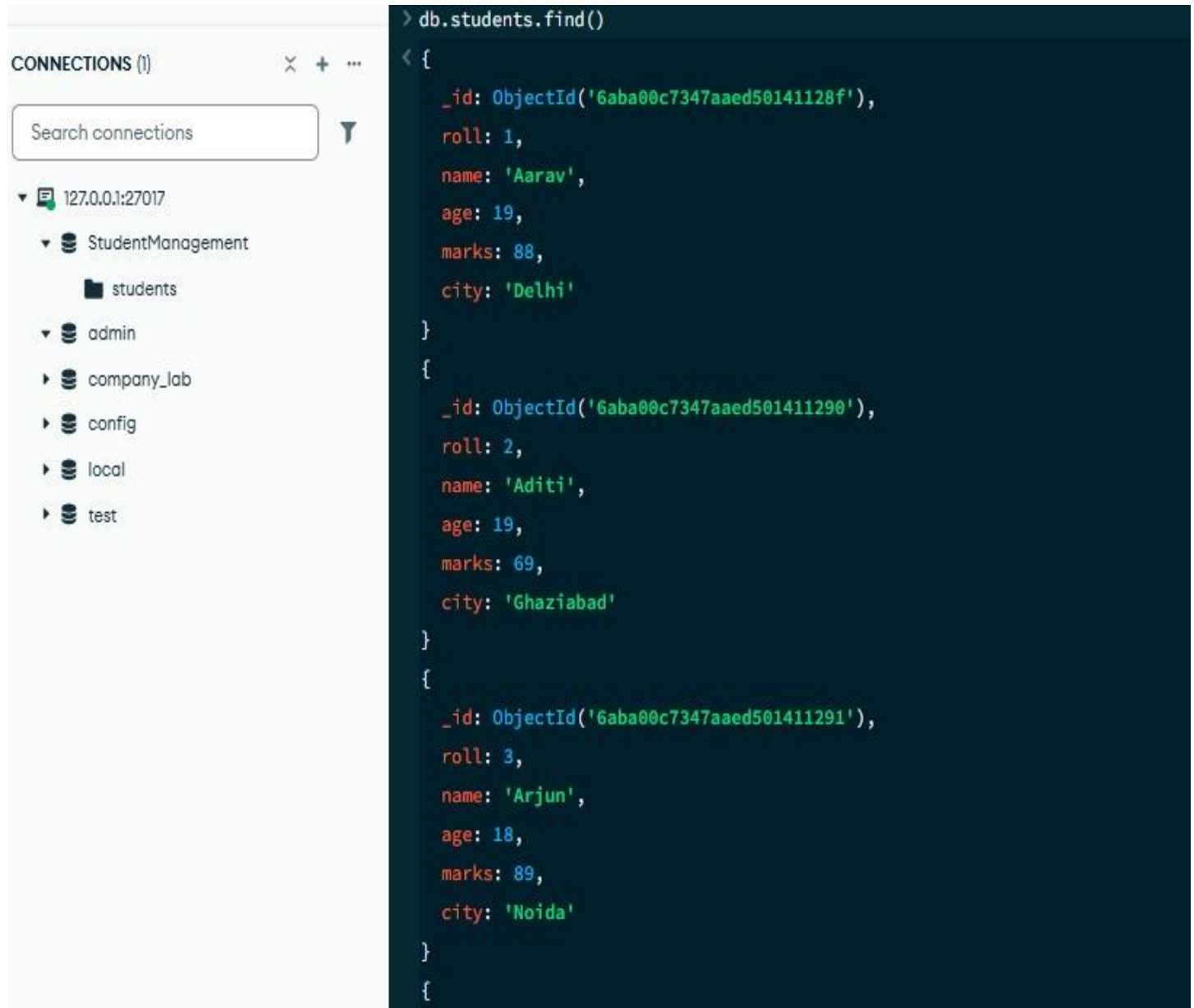
```
> db.students.insertMany([
  {roll:1, name:"Aarav", age:19, marks:88, city:"Delhi"},
  {roll:2, name:"Aditi", age:19, marks:69, city:"Ghaziabad"},
  {roll:3, name:"Arjun", age:18, marks:89, city:"Noida"},
  {roll:4, name:"Bhavya", age:21, marks:57, city:"Delhi"},
  {roll:5, name:"Chirag", age:19, marks:78, city:"Noida"},
  {roll:6, name:"Diya", age:18, marks:90, city:"Lucknow"},
  {roll:7, name:"Dev", age:21, marks:69, city:"Varanasi"},
  {roll:8, name:"Ishita", age:20, marks:55, city:"Ghaziabad"},
  {roll:9, name:"Kunal", age:21, marks:76, city:"Kanpur"},
  {roll:10, name:"Mehak", age:18, marks:64, city:"Delhi"},
  {roll:11, name:"Naman", age:18, marks:60, city:"Prayagraj"},
  {roll:12, name:"Pallavi", age:18, marks:77, city:"Agra"},
  {roll:13, name:"Ritik", age:20, marks:57, city:"Varanasi"},
  {roll:14, name:"Simran", age:18, marks:79, city:"Noida"},
  {roll:15, name:"Tanvi", age:20, marks:95, city:"Bareilly"},
  {roll:16, name:"Varun", age:20, marks:91, city:"Lucknow"},
  {roll:17, name:"Kavya", age:18, marks:57, city:"Lucknow"},
  {roll:18, name:"Raghav", age:20, marks:60, city:"Lucknow"},
  {roll:19, name:"Muskan", age:18, marks:79, city:"Kanpur"},
  {roll:20, name:"Siddharth", age:21, marks:95, city:"Agra"},
  {roll:21, name:"Nikita", age:19, marks:78, city:"Agra"},
  {roll:22, name:"Yash", age:19, marks:97, city:"Kanpur"},
  {roll:23, name:"Ishaan", age:18, marks:93, city:"Ghaziabad"},
  {roll:24, name:"Riya", age:19, marks:65, city:"Varanasi"},
  {roll:25, name:"Mohit", age:21, marks:72, city:"Meerut"},
  {roll:26, name:"Sakshi", age:19, marks:75, city:"Delhi"},
  {roll:27, name:"Ankit", age:19, marks:57, city:"Agra"},
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.students.find()
```

Explanation: Retrieves all documents from the students collection.



The screenshot displays the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel shows a connection to '127.0.0.1:27017'. Under the 'StudentManagement' database, the 'students' collection is selected. The main panel on the right shows the results of the query `db.students.find()`. The results are displayed as a JSON array of three documents, each representing a student.

```
> db.students.find()
< [
  {
    _id: ObjectId('6aba00c7347aaed50141128f'),
    roll: 1,
    name: 'Aarav',
    age: 19,
    marks: 88,
    city: 'Delhi'
  },
  {
    _id: ObjectId('6aba00c7347aaed501411290'),
    roll: 2,
    name: 'Aditi',
    age: 19,
    marks: 69,
    city: 'Ghaziabad'
  },
  {
    _id: ObjectId('6aba00c7347aaed501411291'),
    roll: 3,
    name: 'Arjun',
    age: 18,
    marks: 89,
    city: 'Noida'
  }
]
```

5. EQUAL TO — \$eq

MongoDB Query:

```
db.students.find({marks: {$eq: 88}})
```

Explanation: Finds students whose marks are exactly 88.

```
> db.students.find({marks: {$eq: 88}})
< {
  _id: ObjectId('6aba00c7347aaed50141128f'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed5014112b7'),
  roll: 41,
  name: 'Vivek',
  age: 21,
  marks: 88,
  city: 'Kanpur'
}
{
  _id: ObjectId('6aba01104690e94b489ac173'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.students.find({marks: {$gt: 85}})
```

Explanation: Finds students whose marks are greater than 85.

```
> db.students.find({marks: {$gt: 85}})
< {
  _id: ObjectId('6aba00c7347aaed50141128f'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed501411291'),
  roll: 3,
  name: 'Arjun',
  age: 18,
  marks: 89,
  city: 'Noida'
}
{
  _id: ObjectId('6aba00c7347aaed501411294'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 90,
  city: 'Lucknow'
}
{
```

7. LESS THAN — \$lt

MongoDB Query:

```
db.students.find({marks: {$lt: 60}})
```

Explanation: Finds students whose marks are less than 60.

```
> db.students.find({marks: {$lt: 60}})
< {
  _id: ObjectId('6aba00c7347aaed501411292'),
  roll: 4,
  name: 'Bhavya',
  age: 21,
  marks: 57,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed501411296'),
  roll: 8,
  name: 'Ishita',
  age: 20,
  marks: 55,
  city: 'Ghaziabad'
}
{
  _id: ObjectId('6aba00c7347aaed50141129b'),
  roll: 13,
  name: 'Ritik',
  age: 20,
  marks: 57,
  city: 'Varanasi'
}
{
  _id: ObjectId('6aba00c7347aaed50141129f'),
  roll: 17,
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.students.find({marks: {$gte: 90}})
```

Explanation: Finds students whose marks are 90 or above.

```
> db.students.find({marks: {$gte: 90}})
< {
  _id: ObjectId('6aba00c7347aaed501411294'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 90,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba00c7347aaed50141129d'),
  roll: 15,
  name: 'Tanvi',
  age: 20,
  marks: 95,
  city: 'Bareilly'
}
{
  _id: ObjectId('6aba00c7347aaed50141129e'),
  roll: 16,
  name: 'Varun',
  age: 20,
  marks: 91,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba00c7347aaed5014112a2'),
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.students.find({marks: {$lte: 70}})
```

Explanation: Finds students whose marks are 70 or below.

```
> db.students.find({marks: {$lte: 70}})
< {
  _id: ObjectId('6aba00c7347aaed501411290'),
  roll: 2,
  name: 'Aditi',
  age: 19,
  marks: 69,
  city: 'Ghaziabad'
}
{
  _id: ObjectId('6aba00c7347aaed501411292'),
  roll: 4,
  name: 'Bhavya',
  age: 21,
  marks: 57,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed501411295'),
  roll: 7,
  name: 'Dev',
  age: 21,
  marks: 69,
  city: 'Varanasi'
}
{
  _id: ObjectId('6aba00c7347aaed501411296'),
```



10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(  
  {roll:5},  
  {$set:{marks:95}}  
)
```

Explanation: Updates the marks of the student with roll number 5 to 95.

```
> db.students.updateOne(  
  {roll:5},  
  {$set:{marks:95}}  
)  
  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.students.deleteOne({roll:10})
```

Explanation: Deletes one document matching roll number 10.

```
> db.students.deleteOne({roll:10})  
  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({  
  marks: {$lt:60}  
})
```

Explanation: Deletes all students having marks below 60.

```
> db.students.deleteMany({  
  marks: {$lt:60}  
})  
  
< {  
  acknowledged: true,  
  deletedCount: 12  
}
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions: marks greater than 80 and age 19.

```
> db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
< {
  _id: ObjectId('6aba00c7347aaed50141128f'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed501411293'),
  roll: 5,
  name: 'Chirag',
  age: 19,
  marks: 95,
  city: 'Noida'
}
{
  _id: ObjectId('6aba00c7347aaed5014112a4'),
  roll: 22,
  name: 'Yash',
  age: 19,
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.students.find({
  $or:[
    {city:"Delhi"},
    {city:"Lucknow"}
  ]
})
```

Explanation: Finds students from Delhi or Lucknow.

```
> db.students.find({
  $or:[
    {city:"Delhi"},
    {city:"Lucknow"}
  ]
})
< {
  _id: ObjectId('6aba00c7347aaed50141128f'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed501411294'),
  roll: 6,
  name: 'Diya',
  age: 18,
  marks: 90,
  city: 'Lucknow'
}
```

15A. SORT OPERATION — ASCENDING

MongoDB Query:

```
db.students.find().sort({marks:1})
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

```
> db.students.find().sort({marks:1})
< {
  _id: ObjectId('6aba00c7347aaed501411299'),
  roll: 11,
  name: 'Naman',
  age: 18,
  marks: 60,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6aba00c7347aaed5014112a0'),
  roll: 18,
  name: 'Raghav',
  age: 20,
  marks: 60,
  city: 'Lucknow'
}
{
  _id: ObjectId('6aba01104690e94b489ac17d'),
  roll: 11,
  name: 'Naman',
  age: 18,
  marks: 60,
  city: 'Prayagraj'
}
```

15B. SORT OPERATION — DESCENDING

MongoDB Query:

```
db.students.find().sort({marks:-1})
```

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
> db.students.find().sort({marks:-1})
< {
  _id: ObjectId('6aba00c7347aaed5014112a4'),
  roll: 22,
  name: 'Yash',
  age: 19,
  marks: 97,
  city: 'Kanpur'
}
{
  _id: ObjectId('6aba01104690e94b489ac188'),
  roll: 22,
  name: 'Yash',
  age: 19,
  marks: 97,
  city: 'Kanpur'
}
```

16. LIMIT OPERATION

MongoDB Query:

```
db.students.find().limit(5)
```

Explanation: Displays only the first five documents.

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6aba00c7347aaed50141128f'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed501411290'),
  roll: 2,
  name: 'Aditi',
  age: 19,
  marks: 69,
  city: 'Ghaziabad'
}
{
  _id: ObjectId('6aba00c7347aaed501411291'),
  roll: 3,
  name: 'Arjun',
  age: 18,
  marks: 89,
  city: 'Noida'
}
```

17. SEARCH BY CITY

MongoDB Query:

```
db.students.find({city:"Delhi"})
```

Explanation: Finds all students whose city is Delhi.

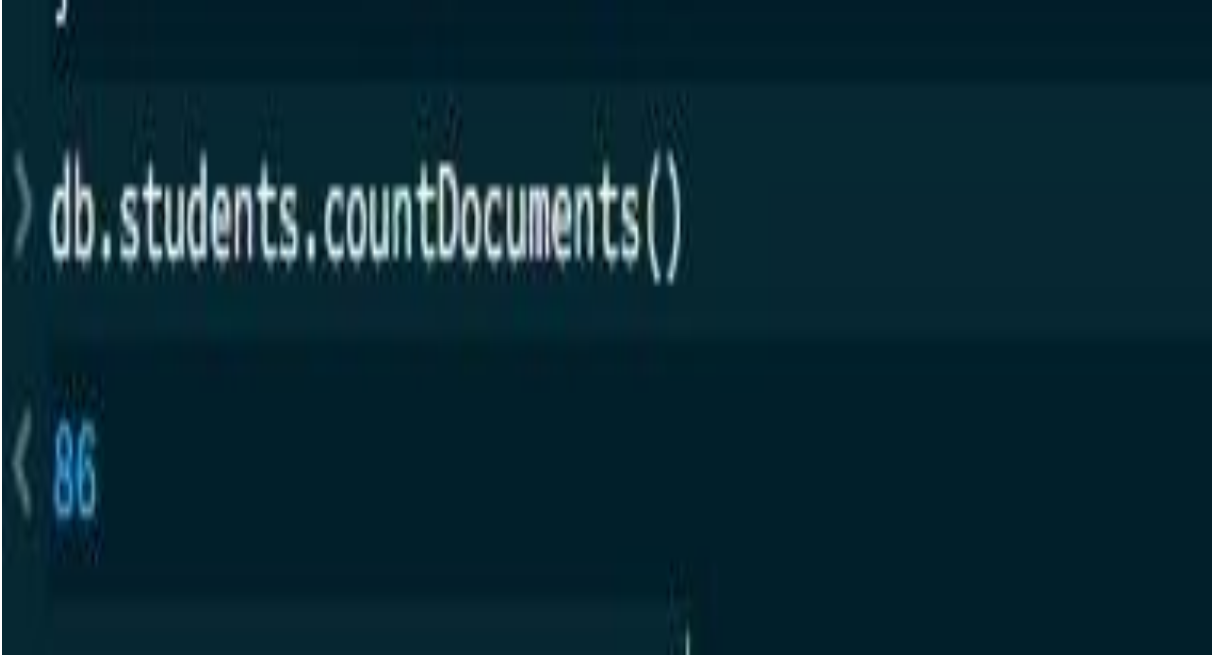
```
> db.students.find({city:"Delhi"})
< {
  _id: ObjectId('6aba00c7347aaed50141128f'),
  roll: 1,
  name: 'Aarav',
  age: 19,
  marks: 88,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed5014112a8'),
  roll: 26,
  name: 'Sakshi',
  age: 19,
  marks: 75,
  city: 'Delhi'
}
{
  _id: ObjectId('6aba00c7347aaed5014112bb'),
  roll: 45,
  name: 'Aditya',
  age: 19,
  marks: 84,
  city: 'Delhi'
}
```

18. COUNT DOCUMENTS

MongoDB Query:

```
db.students.countDocuments()
```

Explanation: Counts the documents currently present in the collection after the delete operations.

A screenshot of a MongoDB shell interface with a dark background. The command `> db.students.countDocuments()` is entered in the prompt. Below it, the result `< 86` is displayed in a light blue color.

MONGODB OPERATORS USED

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:88}
\$gt	Greater than	marks: {\$gt:85}
\$lt	Less than	marks: {\$lt:60}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions must be true	\$and:[...]
\$or	At least one condition must be true	\$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing 50 student records was created, and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to sort data in ascending and descending order, use comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte, and apply logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.